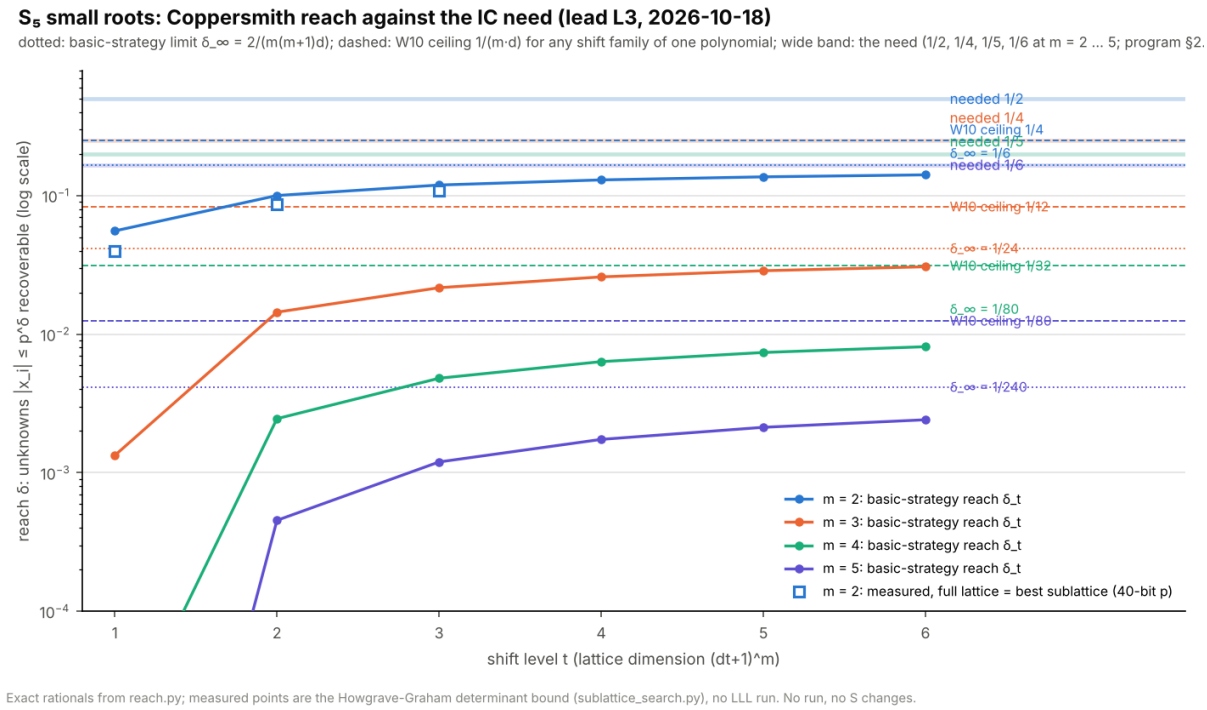
S_5 small roots at $\delta \approx 1/5$: what the uncovered lattice variants can and cannot add (2026-10-18)

Lead L3 of `../ecdlp_exponent_assessment_20261018/LEADS.md`. Class: **closing note, on paper, with one exact computation** no run, no measurement, no claim beyond what the formulas say. The question: the prime-field breakthrough program (`RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md` §2.4, §4.1) needs a decomposition oracle for S_5 that is cheap at $B \approx p^{1/5}$; Probes B and C measured the box-Coppersmith reach, and the W10 study (`research/lit2026_w10_s5_coppersmith_20261011`) proved a family-wide ceiling for every shift family of one polynomial. The program lists variants neither covered (:234–246): Boneh–Durfee sublattices, Herrmann–May unravelled linearisation, cuso Gröbner-optimal shifts for a fixed M, multi-polynomial Coppersmith (adding the curve equations or sharing unknowns across targets), non-box bounds, and the 2026/1027 LP. This note settles which of them the existing bounds already cover.

1. The basic-strategy reach, reproduced exactly



Exact rationals from reach.py; measured points are the Howgrave-Graham determinant bound (sublattice_search.py), no LLL run. No run, no S changes.

`reach_graph.py` draws the table below with the W10 ceilings and the need per `m` (`reach_graph.svg/.png`, values in `reach_graph.diagram.json`).

`reach.py` computes the Jochemsz–May basic-strategy reach on the monomial box `[0, d]^m` of `S_{m+1}(x_1, …, x_m, x_R)` with `x_R` known and every `|x_i| ≤ p^δ`, in exact rationals, and its closed-form limit:

$$\delta_t = (t\cdot\dim - s_p) / s_x, \quad \dim = (dt+1)^m, \quad s_x = \Sigma_{a \in [0,dt]^m} |a|_1,$$
$$s_p = \Sigma_{k=0}^{t} k \cdot |[0,dk]^m \setminus [0,d(k-1)]^m|, \qquad \delta_\infty = 2 / (m\,(m+1)\,d).$$

m	d = 2^{m−1}	needed (§2.4)	δ_1	δ_2	δ_3	δ_∞
2	2	1/2	1/18 = 0.0556	1/10	0.1190	**1/6**
3	4	1/4	0.0013	0.0144	0.0216	**1/24**
4	8	1/5	0.0000	0.0025	0.0048	**1/80**

5	16	1/6	0.0000	0.0005	0.0012	**1/240**

The `m = 2` row is the probes' `1/18, 1/10, 0.119, 1/6` to the digit (`RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md` §B), and `δ_∞ = 2/(m(m+1)d)` is the program's `1/(m(m+1)·2^{m−2})`. Nothing new here; the point is that the formula is now a three-line computation anyone can re-run, and that the level-`t` lattices at `m = 4` are already `6,561`-dimensional at `t = 1` and `390,625` at `t = 3` for a reach of `0.0048`.

2. What each uncovered variant can add

The W10 ceiling: for any full-rank triangular lattice generated by `p^{ℓ−k} x^γ f^k` from the one polynomial (any multiplicity, any monomial set, any leading vertex, the `x`- and `e`-encodings), the reach is bounded by `1/(m·d)` = `1/4, 1/12, 1/32` at `m = 2, 3, 4` (W10 §3, §5.3, §6). W10 §7 is explicit that this does **not** cover non-full-rank sublattice strategies (Boneh–Durfee progressive sublattices, Herrmann–May unravelled linearisation), cuso's out-of-`M` shifts, several polynomials, or non-box bounds. Those were open, and are handled here as follows.

Sublattices, measured. `sublattice_search.py` builds the level-`t` Jochemsz–May basis for the actual S_3 of a 40-bit `a = −3` curve with `x_R` fixed, evaluates the Howgrave-Graham determinant condition exactly (Bareiss on the Gram matrix, so any sublattice can be scored, not only triangular ones), finds the largest `δ` the full lattice passes, and then greedily removes the row whose removal raises `δ` most, until none does.

level `t`	dim	full-rank `δ` (finite `p`, 40 bits)	asymptotic `δ_t` (§1)	best sublattice `δ`	rows dropped
1	9	0.0396	1/18 = 0.0556	0.0396	0
2	25	0.0865	1/10 = 0.1000	0.0865	0
3	49	0.1080	0.1190	0.1080	0

No single-row deletion improves the bound at any of the three levels (`level3.txt`: 49 rows, 0 dropped): the full box lattice is a local optimum under row removal, the box-closure optimality W10 proves for the triangular family seen one step outside it. The finite-`p` values sit below the asymptotic ones by the `√dim` and leading-coefficient terms of the Howgrave-Graham condition, as expected. This is a greedy search, not a proof: a sublattice that pays off only after several simultaneous deletions is not excluded. The magnitudes are the point: published sublattice refinements move a Coppersmith exponent by a few percent (Boneh–Durfee `0.284 → 0.292`; unravelled linearisation gains are of the same size), while this lead needs `1/80 → 1/5`, a factor 16 over the basic strategy and 6.4 over the full-rank ceiling.

variant	what it changes	what binds it here	verdict
Boneh–Durfee sublattice selection	drops rows to lower `det`	no ceiling proved; greedy removal finds no gain at `t ≤ 2`; literature gains are percent-level	cannot plausibly supply 6.4–16×
Herrmann–May unravelled linearisation	relinks products of unknowns	same lattice class as the `e`-encoding, which W10 shows does no better	≤ `1/(md)` where triangular; percent-level otherwise

cuso Gröbner-optimal shifts at fixed M	shifts whose support leaves M	W10 exact optimum for ℓ ≤ 3 inside the box; out-of-M shifts unbounded by W10	the box is closure-optimal; a one-row gain would show in the greedy search and none appears
non-box bounds (`	x_i	≤ p^{δ_i}` unequal)	reweights s_x
the 2026/1027 LP	an LP bound on the asymptotic reach	would tighten, never loosen, the ceiling	can only lower the numbers above

3. Multi-polynomial systems: what the IC setting does and does not supply

Coppersmith's reach rises when several *independent* modular equations share the same small unknowns (with k such equations the p-exponent sum s_p scales roughly with k, so the reach can approach k·δ_1). To lift 1/80 to 1/5 at m = 4 would take the equivalent of 16 independent degree-8 relations in the same four abscissae. The relation phase of index calculus supplies, per decomposition attempt, exactly one: $S_5(x_1, \ldots, x_4, x_R) \equiv 0$.

- **Adding the curve equations** $y_i^2 = x_i^3 + a\ x_i + b$ introduces one full-size unknown y_i per summand (nothing bounds y_i by p^δ), so each added equation costs more lattice volume than it contributes; the summation polynomial already eliminated the y_i for this reason.
- **Sharing unknowns across targets** needs two targets R, R' whose decompositions use a common factor-base point; that is unknown before the decomposition succeeds, and over random targets the expected number of shared summands among m is m^2/B, negligible at B = p^{1/5}.
- **Several summation polynomials for one target** (S_5 at different m-subsets, or S_3 chains with auxiliary points) introduce the auxiliary abscissae as new unknowns; the Koblitz chaining count m·ℓ + (m−2)·n is the same bookkeeping in that setting, and it grows, it does not shrink, the unknown count.

None of the three supplies a second independent equation in the same small unknowns, so the multi-polynomial route has no input to work on in this setting. This is the honest reading, not a theorem: a construction that produced several independent low-degree relations in the same m small abscissae from one target would be exactly the new embedding the program's decision (§5) asks for before any further lattice run.

4. Verdict

- Lead L3 is **closed for the triangular family by W10's theorem** (1/32 < 1/5) and **closed by magnitude, not by theorem, for the sublattice variants**: the greedy determinant search finds no gain at the probes' levels, and the literature's sublattice gains are percent-level against a 6.4–16× need.
- The multi-polynomial variants are **not applicable** as stated: the setting provides one equation per decomposition and no shared small unknowns.
- What would re-open it: an embedding that yields k ≥ 16 independent modular relations of degree ≤ 8 in the same four factor-base abscissae per target, or a structured factor base whose membership adds such relations at degree ≪ B (which KN-OPEN-020's bounded-degree theorem forbids for bounded-degree predicates). Either would be tested with the Probe B/C framework unchanged.

Cost of this note: reach.py runs in under a second; sublattice_search.py runs in minutes at levels 1–2 and under half an hour at level 3; no lattice was reduced (the determinant bound is what every reach number quotes). Sources: RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md §2.2, §2.4, §3.2–

3.3, §4.1; research/lit2026_w10_s5_coppersmith_20261011/README.md §3, §5, §6, §7; KN-OPEN-020 (crypto-autoresearcher).